PERSONAL CAPSTONE PROJECT

SUPERMART PRODUCT PRICE ANALYSIS

Web Scraping, Data Cleaning,
Exploratory Data Analysis and
Power BI Dashboard Development

FINAL TECHNICAL REPORT

Executive Summary

This project shows the main steps I followed in data analysis: collecting online supermarket product listings, cleaning the extracted data, exploring price distributions, and communicating findings through an interactive Power BI report. The source website was Supermart.ng. Four categories were examined: Drinks, Fresh Food, Food Cupboard and Snacks.

My final cleaned file contains 1,100 product listings and three fields: Product Name, Price and Category. Python Requests and BeautifulSoup were used in the scraping workflow; Pandas supported data cleaning and exploratory analysis; Matplotlib supported the box plot; and Power BI provided the dashboard experience.

Verified headline results

Indicator	Finding
Products	1,100
Overall mean price	₦7,810.46
Median price	₦2,877.50
Most frequent price	₦365.00
Minimum / maximum	₦90 / ₦980,000
IQR high-price outliers	88 listings (8.0%)

Fresh Food had the highest average (₦10,709.78), while Snacks had the lowest (₦3,882.30).

The substantial gap between the overall mean and median suggests a distribution with a long tail of high prices: a relatively small number of very expensive listings raise the average. The project is descriptive rather than causal; prices alone cannot establish consumer demand, revenue, sales volume or profitability.

The Power BI report includes three principal analytical pages—Product Overview, Price Analysis and Additional Analysis—alongside pre-analysis and insights material. The report is designed to document the original work and the interpretation of its results.

Currency note: N denotes Nigerian naira (NGN) throughout this report.

1. Introduction and Background

1.1 Context

Online retail platforms publish product listings with information that can be collected and analysed systematically. It is difficult to understand hundreds of prices by looking at them one by one: individual price tags do not immediately reveal a typical price, the balance between inexpensive and premium items, or differences between product categories. Data analytics converts these separate listings into a structured view.

This project uses Supermart.ng as a practical source of online supermarket product data. It follows a common analytics sequence: obtain records, standardise them, assess data quality, compute descriptive measures, visualise the distribution, and design a report that allows users to explore the results interactively.

1.2 Project rationale

The dataset was chosen because it supports real-world practice in both programming and business-oriented reporting. Product Name and Category are descriptive attributes, while Price is a measurable numeric attribute. Together, they support grouping, comparison, ranking, frequency distributions and finding unusually high or low values.

The purpose is not to claim that one category is intrinsically more expensive than another in every supermarket. Instead, the project documents how prices differ within the listings collected. This distinction matters because categories may contain different types, brands, pack sizes and product mixes.

1.3 Analytical framing

For the pre-analysis board, Category is the principal grouping or explanatory variable and Price is the numerical outcome being compared. Product Name is an identifier/description. These are analytical roles, not evidence of causation. The project does not contain dates, quantities sold, costs or customer transactions, so no time trend, demand model or profitability estimate is supported.

2. Problem, Aim and Research Questions

2.1 Problem statement

Reviewing a large online product catalogue manually makes it difficult to compare typical prices and detect unusual values consistently. A reproducible data workflow is needed to bring the listings into a single table, ensure prices are numeric, and present interpretable summaries.

2.2 Aim

To collect and analyse selected Supermart.ng product listings and develop interactive visualisations that communicate price patterns across four supermarket categories.

2.3 Specific objectives

• Collect at least 1,000 product listings using Python web scraping.

• Combine four categories into a consistent dataset with Product Name, Price and Category.

• Check missing values, duplicate records and numeric price conversion.

• Calculate mean, median, mode, minimum, maximum and category summaries.

• Identify unusually high prices using the interquartile-range (IQR) rule.

• Build at least seven distinct visualisation types across Python and Power BI.

• Develop filterable Power BI pages with summary cards and navigation.

2.4 Research questions

• How many listings are represented in each category?

• What are the overall average, median and modal prices?

• Which category has the highest and lowest average price?

· How many listings fall into the selected price ranges?

• What are the extreme prices and the spread of the data?

• Which categories account for unusually high-priced listings?

• How can interactive visuals help readers compare these findings?

3. Data Source and Collection Design

3.1 Source and sampling scope

The source was the public product listing pages of Supermart.ng. Four collection paths were used: /collections/drinks, /collections/fresh-food, /collections/food-cupboard and /collections/snacks-treats. The original workflow covered ten pages each for Drinks and Fresh Food, and one page each for Food Cupboard and Snacks.

This was a selected-category, selected-page collection rather than a random sample of the whole website. The category counts therefore reflect the chosen collection design and should not be interpreted as the true distribution of all products available on Supermart.ng.

Category Pages in collection Rows

Drinks 10 500

Fresh Food 10 500

Food Cupboard 1 50

Snacks 1 50

Total 22 page visits 1,100

3.2 Original fields

• Product Name: listing title extracted from product-card HTML.

• Price: displayed naira price, originally obtained as formatted text.

• Category: collection label assigned during extraction.

3.3 Important scope conditions

Listings may represent different package sizes or multipacks. Consequently, a higher listed price does not necessarily imply a higher price per unit. The file does not record scraping timestamps for individual products or historic price changes, so it supports a cross-sectional analysis only.

4. Web Scraping Methodology

4.1 Python libraries and request process

I used the Requests library in Python to retrieve HTML from the selected collection pages. BeautifulSoup parsed that HTML so that product information could be selected from the page structure. Pandas converted extracted records into DataFrames, making them suitable for validation, combination and export.

The observed product selectors in the project workflow included p.card__title with a product link and span.price__current for the displayed price. Correctly pairing each product title with its corresponding price is important: selecting unrelated lists of names and prices could silently produce incorrect records.

4.2 Pagination and category labelling

The Drinks and Fresh Food collections were gathered across multiple pages, using page-number parameters in the collection URLs. Each record received the appropriate category label. A category-path issue encountered during development was resolved by using /collections/fresh-food rather than the shorter /collections/food path.

The notebook saved intermediate category CSVs and combined them using Pandas concatenation. The combined output was then cleaned and saved as supermart_cleaned.csv. The original notebook is the primary evidence of the workflow; the PDF does not require rerunning website requests, which could return changed prices or product availability.

4.3 Reliability and responsible collection

A scraping workflow should check HTTP status codes, ensure the intended number of product cards was found, inspect sample rows and compare totals after pagination. Collection should respect website access policies and avoid unnecessary repeated requests. Since product catalogues change, the saved dataset—not a new live scrape—is the reproducible basis for the present analysis.

5. Data Cleaning and Quality Assurance

5.1 Consolidation

The four category DataFrames were combined into one three-column table. The resulting 1,100 rows were checked for completeness and duplicate records. A clean table is necessary because a non-numeric Price field would prevent accurate averages, quartiles and numerical comparisons.

5.2 Price transformation

Displayed prices contained the naira symbol and thousands separators, e.g. NGN 5,880.00.

characters were removed and the resulting strings were converted with Pandas to a numeric price type.

The final CSV stores Price as a floating-point number, allowing arithmetic calculations.

After converting the Price column, I checked its data type to confirm that the values were numeric and ready for analysis.

5.3 Validation results

Quality test	Result
Row count	1,100
Columns	Product Name, Price, Category
Missing values	0 in every column
Exact duplicate rows	0
Distinct product names	1,099
Price type	float64
Categories	4

One product name—Farm Fresh Yoghurt Banana 1 L—appears in both Drinks and Fresh Food at the same listed price. The category labels differ, so the rows are not exact duplicates. Removing one automatically could discard meaningful collection information; this is best disclosed as a cross-category listing overlap.

6. Exploratory Data Analysis: Overall Prices

6.1 Descriptive statistics

Statistic Verified value

Number of listings 1,100

Mean ₦7,810.46

Median ₦2,877.50

Mode ₦365.00

Minimum ₦90.00

First quartile (Q1) ₦990.00

Third quartile (Q3) ₦6,623.75

Maximum ₦980,000.00

Standard deviation ₦36,728.52

6.2 Interpretation

The average is more than twice the median. This indicates that relatively high values exert substantial upward pressure on the mean. The median is therefore a useful different measure of a typical listing, especially when comparing categories with very different spreads.

The mode, ₦365, is the most common exact price in the data, but it does not imply that most listings are priced near ₦365. Exact price frequencies are different from broad price ranges, and the distribution includes both inexpensive items and exceptionally high-priced listings.

6.3 Correlation limitation

The original dataset has only one numeric field: Price. A conventional correlation coefficient between two independent original numeric measures is therefore not meaningful here. Product Count is a derived aggregation, not a second raw numerical observation for each product. The analysis appropriately focuses on descriptive statistics, category comparisons and distributions instead of inventing a price-versus-sales correlation.

7. Category-Level Comparisons

Figure 1. Arithmetic mean listed price by product category, recalculated from the cleaned CSV.

Category Count Mean (N) Median (N)

Drinks 500 5,516.21 2,980.00

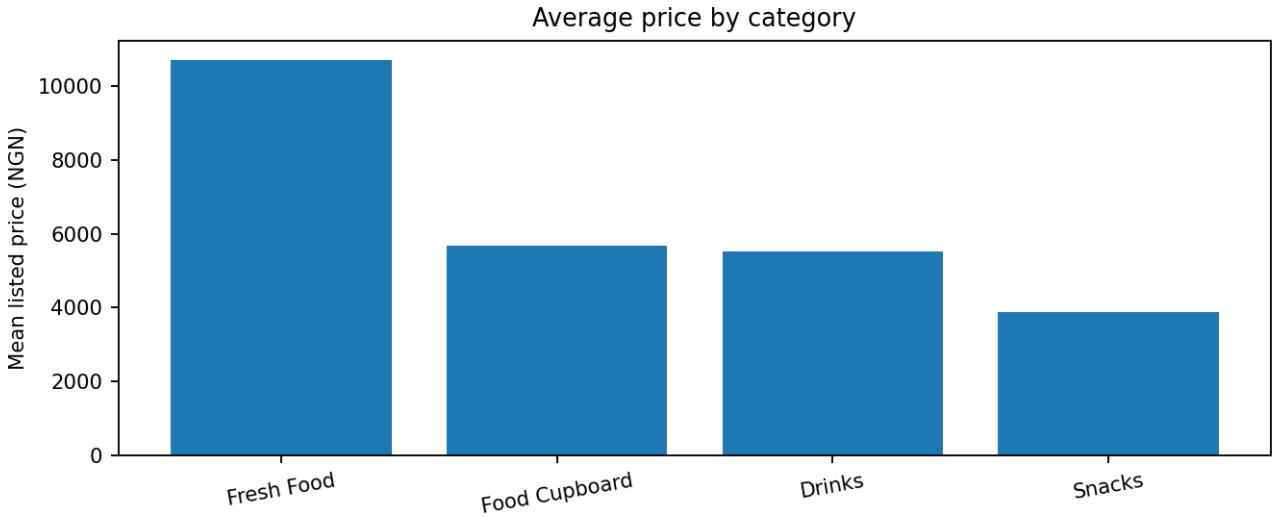
Fresh Food 500 10,709.78 2,627.50

Food Cupboard 50 5,688.00 3,975.00

Snacks 50 3,882.30 2,465.00

Fresh Food has the highest mean, but its median (₦2,627.50) is below the Drinks median (N 2,980.00). This is a useful example of why mean and median should be reported together: the Fresh Food average is heavily affected by expensive listings, including the overall maximum.

Food Cupboard has a slightly higher mean than Drinks, but the categories contain only 50 and 500 records respectively. This uneven sampling should be considered when making comparisons; the chart describes the observed sample rather than proving a general pricing hierarchy.



8. Price Ranges and Product Distribution

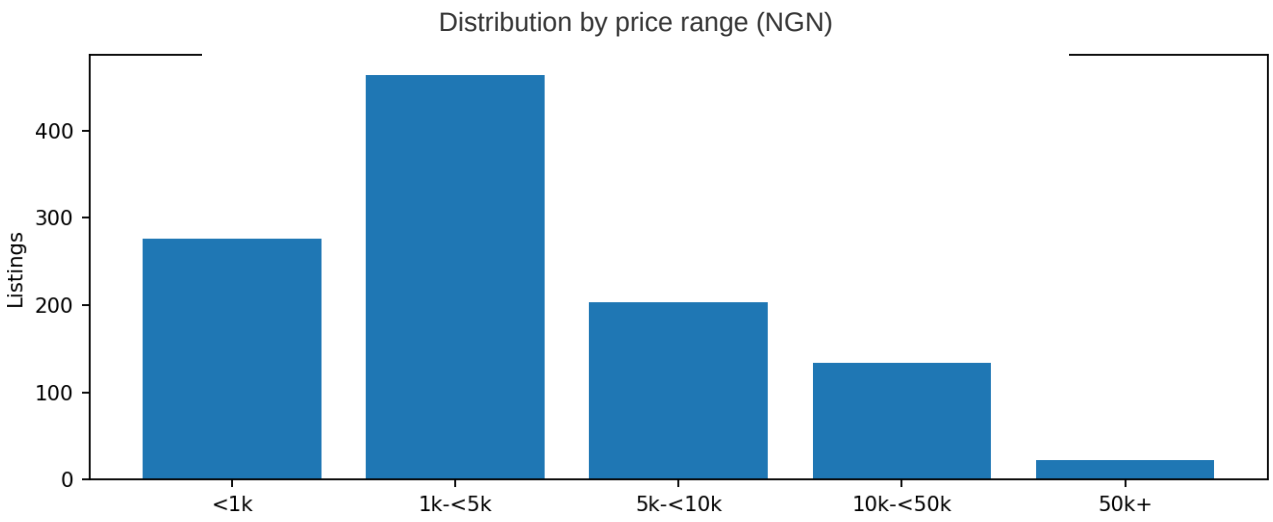
Figure 2. Product count by price range; intervals are left-inclusive and right-exclusive.

Price range	Count	Share
Below ₦1,000	276	25.1%
₦1,000 to below 5,000	464	42.2%
₦5,000 to below 10,000	203	18.5%
₦10,000 to below 50,000	134	12.2%
₦50,000 and above	23	2.1%

The largest group is the ₦1,000–4,999 range, containing 464 records. Together with the 276 listings below ₦1,000, this means 740 of the 1,100 records (67.3%) are priced below ₦5,000.

Only 23 listings are priced at ₦50,000 or more. Their small frequency does not make them analytically unimportant: they can materially affect the overall mean, chart axes and comparisons between categories.

The displayed counts use non-overlapping intervals. A price of exactly ₦1,000 belongs to the second range; a price of exactly ₦5,000 belongs to the third. This avoids double-counting at range boundaries.



9. Outlier Detection and Box Plot

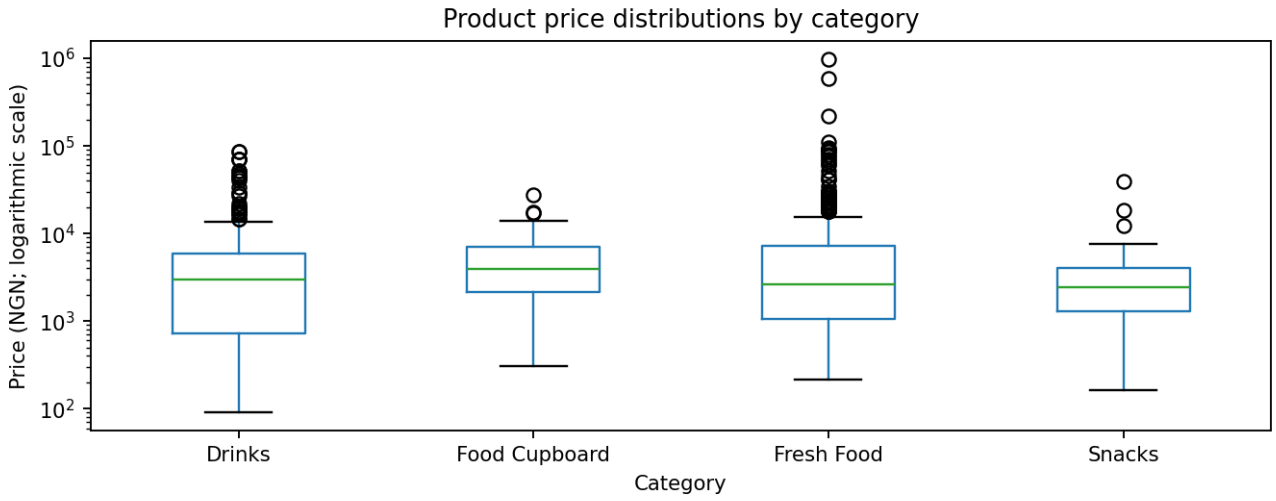
9.1 Interquartile-range method

The IQR measures the spread of the middle half of the data. For this dataset, Q1 = ₦990.00 and Q3 = ₦6,623.75, giving an IQR of ₦5,633.75. The conventional upper fence is Q3 + 1.5 × IQR = N 15,074.375. The lower fence is Q1 − 1.5 × IQR = −₦7,460.625.

Using these boundaries, 88 listings are high-price outliers (8.0% of the dataset), and none fall below the lower fence. The term “outlier” describes a statistical extreme, not necessarily an error: some high prices may be genuine premium or bulk products.

Figure 3. Box plot of Price by Category on a logarithmic vertical axis.

The log scale helps reveal category distributions despite the exceptionally high maximum. It changes how distances appear on the chart but does not change the underlying data or the IQR rule. High-price outliers comprise 53 Fresh Food, 30 Drinks, 3 Food Cupboard and 2 Snacks records.



10. Visualization Design and Rationale

The capstone requires at least seven different visualization types. The combined Python and Power BI work uses the following distinct types:

Type Analytical purpose

Column chart Compare average price across categories

Horizontal bar chart Compare category listing counts

Doughnut chart Show category shares of the selected measure

Treemap Compare price-range breakdown

Matrix Display category-level numerical summaries

Scatter plot Explore plotted average price and product count

Waterfall chart Display cumulative contribution of category counts

Box plot (Python) Summarise spread, quartiles and outliers

Design principles

Chart selection follows the question being answered. Counts are easiest to compare with bars; averages with columns or tables; distributions with price ranges and box plots; and multi-measure exploration with a scatter plot. The Power BI pages use coordinated colours, white visual panels and a dark-blue background to separate sections visually.

A visual title must agree with the measure shown on its axes. I supplied Power BI screenshots explicitly showing “Average of Price” for the average-price column chart and matrix. The scatter plot was described as Average Price against Product Count. The final report should preserve those labels and avoid confusing averages with sums.

Interpretation caution

A chart can look convincing even when it depicts a derived or aggregated measure. In this report, “product count” means number of collected listings; it is not units sold. A “category share” is likewise a share of the chosen aggregation and should not be called market share or revenue share.

11. Power BI Dashboard: Product Overview

Figure 4. Power BI screenshot showing the Product Overview visual fields and charts.

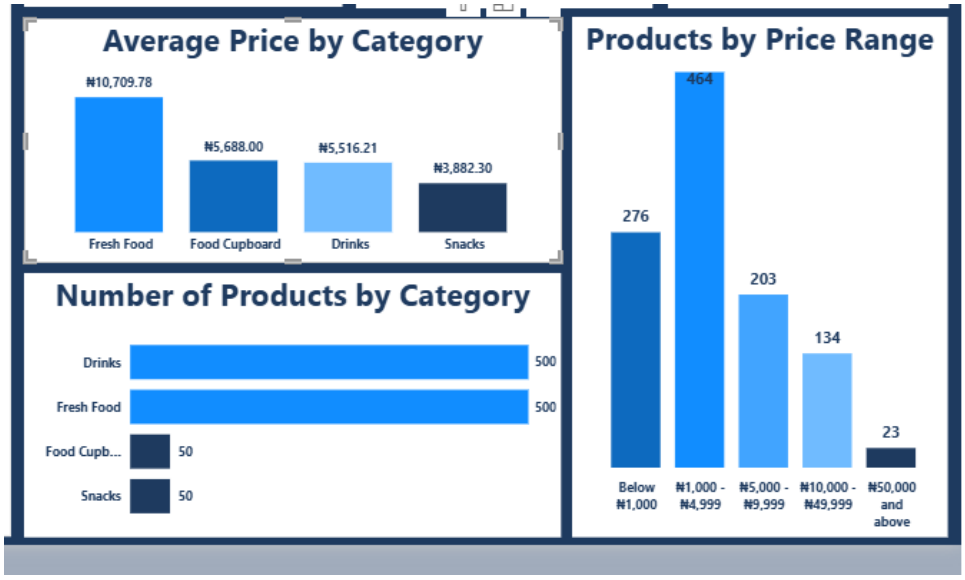
Page purpose and components

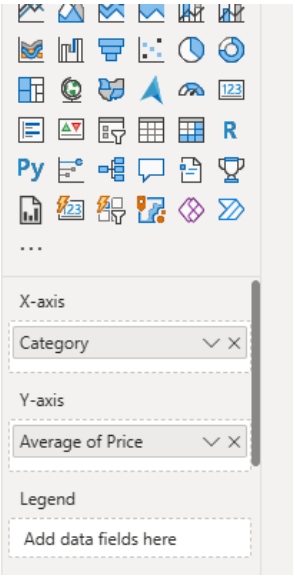
The Product Overview page introduces the dataset through a category-average column chart, a horizontal category-count chart and a price-range column chart. Together these visuals answer three foundational questions: which category has the highest mean price, how many records were collected per category, and where most prices lie.

The Average Price by Category chart displays Fresh Food at ₦10,709.78, Food Cupboard at N 5,688.00, Drinks at ₦5,516.21 and Snacks at ₦3,882.30. The Power BI field settings show the Y-axis uses Average of Price. The category-count chart shows 500, 500, 50 and 50 records, matching the CSV.

Interaction

I tested the category and price-range slicers and the page-navigation buttons in Power BI Desktop. The slicers filter the displayed data, while the buttons allow movement between dashboard pages. The interactive features are available in the Power BI file.





12. Power BI: Price and Additional Analysis

12.1 Price Analysis page

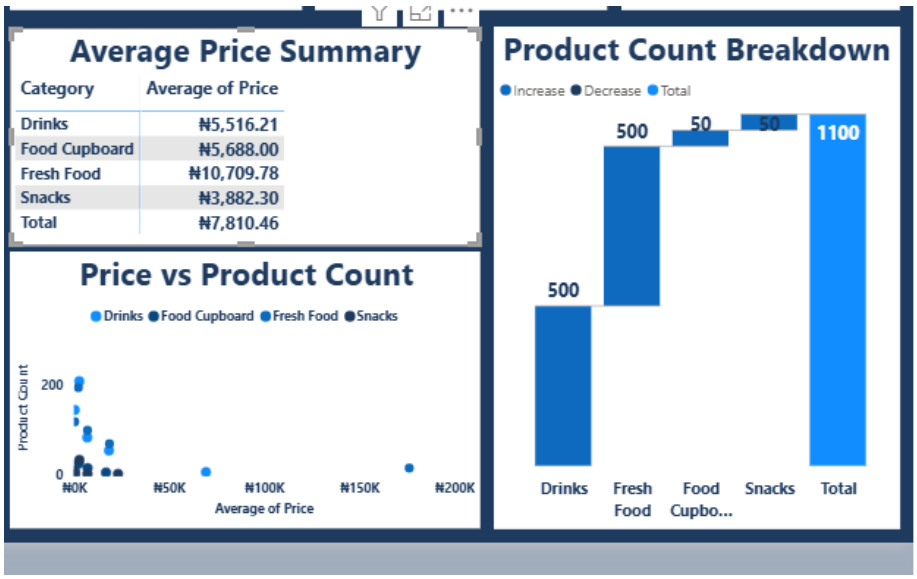
The Price Analysis page uses a doughnut chart for category share and a treemap for price-range breakdown. These provide different ways of viewing breakdown: the doughnut emphasises proportions, while the treemap emphasises relative areas. Both should be interpreted using the measure assigned to each visual, not as evidence of sales or market share.

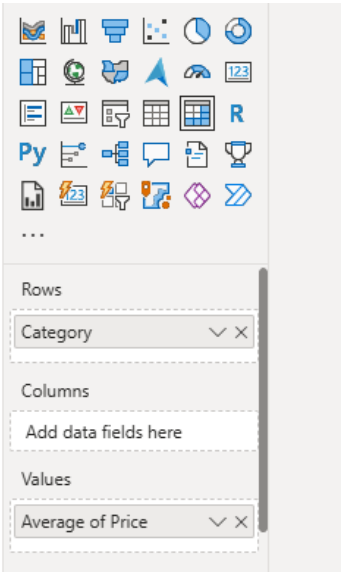
12.2 Additional Analysis page

Figure 5. Power BI Desktop screenshot of the matrix, scatter plot and waterfall chart.

The Average Price Summary matrix lists category means and an overall mean of ₦7,810.46. The supplied screenshot confirms its Values field is Average of Price. The scatter plot displays Average Price on its horizontal axis and Product Count on its vertical axis. The waterfall chart builds the 1,100 total from the four category contributions: 500, 500, 50 and 50.

I tested the dashboard slicers and navigation buttons in Power BI Desktop.
The slicers and navigation buttons worked correctly during testing.





13. Findings, Insights and Interpretation

Finding 1 — Distribution is uneven, with some very high prices

The overall mean of ₦7,810.46 is substantially higher than the median of ₦2,877.50. This indicates that relatively expensive listings have an outsized effect on the average. A report relying only on the mean would obscure the experience of the middle-priced listing.

Finding 2 — Fresh Food has the highest mean

Fresh Food leads on mean listed price (₦10,709.78), yet its median is ₦2,627.50. The contrast suggests substantial within-category variation rather than uniformly high prices. The maximum of N 980,000 warrants careful inspection of the associated listing and its unit or pack description before drawing strong pricing conclusions.

Finding 3 — Most listings are below ₦5,000

The two lowest price ranges contain 740 listings, or 67.3% of the data. Thus, although a few extreme listings raise the overall average, the majority of collected products fall below ₦5,000.

Finding 4 — Outliers cluster in larger categories

Of the 88 IQR high-price outliers, 53 are Fresh Food and 30 are Drinks. These categories also account for 1,000 of the 1,100 listings, so absolute outlier counts should not be interpreted without considering the category denominators.

Finding 5 — Sampling affects comparisons

Drinks and Fresh Food each contribute 500 records, while Food Cupboard and Snacks contribute only 50 each. Overall statistics and category comparisons are consequently influenced by both the product mix and the collection design. The results are most defensible as an analysis of this saved sample.

14. Recommendations, Limitations and Improvements

14.1 Recommendations

• Report both mean and median. Their difference is a central finding and helps readers avoid mistaking the mean for a typical listing.

• Investigate extreme listings. Inspect the ₦980,000 maximum and other high-price outliers for product type, pack size and extraction accuracy before any operational interpretation.

• Use filters when comparing categories. Slicers and category summaries support focused exploration without changing the original records.

• Preserve the cleaned dataset and notebook. Keeping the CSV, Python code and PBIX together improves traceability and reproducibility.

• Document sampling differences. The 500/500/50/50 distribution must accompany cross-category claims.

14.2 Limitations

The dataset covers only four selected categories and a limited set of listing pages. It records displayed prices, not verified checkout prices, historical trends or actual transactions. Product sizes are not standardised into comparable unit prices. Some product names can occur across categories. The dataset also contains only one original numerical variable, preventing a meaningful pairwise numerical correlation analysis.

14.3 Suggested future improvements

A later version could capture scrape timestamps, brand, package quantity, unit of measure, product URL and stock status. Standardising price per kilogram, litre or item would improve comparisons. Repeated collections over time could support trend analysis; verified sales data, if available, could support demand-related questions. None of those analyses is claimed as completed in this capstone.

15. Conclusion

15.1 Summary of the project

In this project, I collected and analysed 1,100 Supermart.ng product listings from Drinks, Fresh Food, Food Cupboard and Snacks. I used Python and BeautifulSoup for web scraping, Pandas for cleaning and exploratory analysis, and Power BI to present the findings in interactive dashboards.

The final dataset had no missing values or exact duplicate rows. The average listed price was ₦7,810.46, while the median was ₦2,877.50. Fresh Food had the highest category average, and Snacks had the lowest. Most listings were priced below ₦5,000, and the IQR method identified 88 high-price outliers.

15.2 What the results mean

These results show why it is important to compare both average and median prices. A small number of expensive products can raise the average and make it less representative of a typical listing. The dashboards make the differences between categories and price ranges easier to understand.

15.3 Limitations and future improvements

This analysis is based on product listing prices, not completed sales or customer demand. The categories also contain different numbers of listings, and products may differ in size or quantity. In future work, I would collect data across more dates, standardise comparable product sizes, and expand the smaller categories to support stronger comparisons.

Overall, this capstone strengthened my practical skills in collecting, preparing, analysing and communicating data.